MILITARY SPECIFICATION

# TRANSMITTING SET, TELEMETRY DATA AN/AKT-21

This Military Specification has been approved by the Naval Air Systems Command, Department of the Navy

## 1. SCOPE

1.1 Scope.- This specification establishes the general design and performance requirements for a telemetry transmitting set intended for use in QT-33 or BMQ-34A & E targets in order to provide data facilitating remote control of these targets. The telemetry transmitting set shall be of the FM/FM type and shall contain 10 channels.

1.2 Classification.- The telemetry transmitting set shall consist of the following:

| Item | Applicable Paragraph |
|---|---|
| Telemetry Transmitting Set | 3.4.1 |

## 2. APPLICABLE DOCUMENTS

2.1 General.- The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-172 | Cases; Bases, Mounting, and Mounts |
| MIL-W-5088 | Wiring; Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-T-5422 | Testing Environmental Aircraft Electronic Equipment |



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SPECIFICATIONS

Military

| | |
|---|---|
| MIL-I-6181 | Interference Control Requirements Aircraft Equipment |
| MIL-E-17555 | Preparation for Delivery of Electronic, Miscellaneous Electrical (Except Rotating Electrical Equipment), and Associated Repair Parts |
| MIL-D-18300 | Design Data Requirements for Avionics Equipment |
| MIL-T-18303 | Test Procedures, Preproduction and Inspection, for Aircraft and Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Airborne Electronic and Associated Equipment |
| MIL-R-22973 | Reliability Index Determination for Avionic Equipment Models, General Specification for |
| MIL-T-23336 | Technical Directives (Letter Type), Preparation of |
| MIL-STD-422 | Telemetry Standards for Missiles and Aircraft |
| MIL-STD-470 | Maintainability Program Requirements |
| MIL-STD-471 | Maintainability Demonstration |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MIL-STD-781 | Reliability Tests |
| MIL-STD-785 | Requirements for Reliability Programs |
| MIL-STD-1304 | Reliability Reporting |



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2.2 Availability of Documents.- When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Parts and Materials.- In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and material requirements shall conform to specification MIL-E-5400.

(b) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-between-failures (MTBF) of more than 5,000 hours should be the goal of each.

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval.- Approval for the use of nonstandard parts and materials shall be obtained as outlined in specification MIL-D-18300. (See 6.6.5)

3.2 Design and Construction.-

3.2.1 General.- The equipment shall conform with all the applicable requirements of specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.1.1 Design Data.- No data is required by this specification, or by applicable documents referenced in section 2 unless specified in the contract or order. (See 6.6.1)



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3.2.2 Total Weight.- The total weight of the equipment excluding cables, shall be a minimum consistent with good design, and shall not exceed 13 pounds.

3.2.3 Operating Life.-

3.2.3.1 Total Operating Life.- The equipment shall operate within the requirements of this specification subject to only limited servicing. The operating life of the equipment, after acceptance by the Government, shall be at least 270 hours of accumulated on-time at the extreme environmental conditions specified.

3.2.4 Reliability.-

3.2.4.1 Reliability Program.- A reliability program shall be conducted in accordance with MIL-STD-785. Reliability reporting shall be in accordance with MIL-STD-1304.

3.2.4.2 Mean-Time-Between-Failure.- The equipment shall have a Mean Time Between Failures (MTBF) of at least 19 hours under field test level conditions in accordance with Military Specification MIL-R-22973. This is based on a 90% probability of success for a two hour mission. The test level conditions shall be based upon temperature extremes of -54°C to +55°C and the vibration requirements of MIL-E-5400, curve I, figure 2. The specified MTBF shall be demonstrated in the testing of one equipment at a lower confidence level of 90% in accordance with MIL-R-22973. The length of the test shall be such that accumulated total equipment on time shall be 270 hrs. under the conditions of paragraph 3.3 of MIL-R-22973.

A failure is defined as any malfunction which causes the equipment to fail to meet the performance requirements of this specification.

3.2.5 Cabling and Connections.-

3.2.5.1 Cables and Connectors.- The equipment shall provide for the use of cables and connectors in accordance with specification MIL-E-5400.

3.2.5.2 Interconnection Cabling.- The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables shall not be supplied; however, the external mating connectors shall be supplied as part of the equipment.



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3.2.5.3 Input Power Connection.- Input power connections shall be in accordance with MS 25245.

3.2.6 Interchangeability.- Electrical, mechanical and functional interchangeability is required between assembled units delivered under the pertinent contract and previously produced units. All parts, subassemblies, assemblies and units produced under the pertinent contract shall be electrically, mechanically and functionally interchangeable with like items delivered under the pertinent contract.

3.2.7 Interference Control.- The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of specification MIL-I-6181.

3.2.8 Maintainability Provisions.- Shall conform to the requirements of MIL-STD-470 as concerns corrective maintenance only, and shall be based on the replaceable assembly concept. (M max CT) shall be less than one hour.

3.2.9 Nomenclature and Nameplates.- Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with specification MIL-N-18307. (See 6.6.6)

3.2.10 Standard Conditions.- The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

| | |
|---|---|
| Temperature | Room Ambient (30° ±10°C) |
| Altitude | Normal Ground |
| Vibration | None |
| Humidity | Room Ambient up to 90 Per Cent Relative Humidity |
| Input Power Voltage | 28.0 ±10% vdc |

3.2.11 Service Conditions.- The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in specification MIL-E-5400 for Class 2 equipment as follows:

3.2.11.1 Operating Temperature.- Per Table I, Column I

3.2.11.2 Nonoperating Temperature.- Per Table I, Column IX except that the lowest nonoperating temperature is -54°C instead of -62°C.



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3.2.11.3 Altitude.- Per Table I, Column VIII.

3.2.11.4 Temperature-Altitude Combination.- Per Table I, Column IV.

3.2.11.5 Vibration.- Per Curve I of Figure 2 with no vibration isolators.

3.2.11.6 Humidity.- Per paragraph 3.2.24.4. Note that proof testing in accordance with MIL-T-5422 shall be for 5 cycles instead of 10 cycles.

3.2.11.7 Shock.- Per paragraph 3.2.24.6.1.

3.2.11.8 Explosive Conditions.- Per paragraph 3.2.24.10

3.2.11.9 Other Requirements.- Sand and dust per paragraph 3.2.24.7, Fungus per paragraph 3.2.24.8 and salt atmosphere per paragraph 3.2.24.9. Sand and dust, fungus and salt spray testing are not required.

3.2.12 Warmup Time.- The time required for the equipment to warmup prior to operations shall be kept to a minimum, and shall not exceed 1 minute under standard conditions, 5 minutes after 2 hours non-operating at -20°C, and 20 minutes after non-operating at -54°C.

3.2.13 Input Power Requirements.- The equipment shall meet the requirements of specification MIL-STD-704, paragraphs 5.2.2.1 and 5.2.3, and shall give specified performance from the following power source with characteristics as defined in specification MIL-STD-704, and having limits as specified herein. The power shall not exceed the specified amounts. Note that the Audio Susceptibility requirements of MIL-E-6181 exceed those of MIL-STD-704. Both requirements pertain.

(a) D.C. power, 28 volts ±10%, 8 amperes.

3.2.13.1 Grounded Input.- The negative of the D.C. power source may be grounded externally with no adverse effects on the equipment.

3.2.13.2 Undervoltage Protection.- The equipment shall not be damaged by voltages below the minimum specified herein, and shall automatically resume normal operation when the voltage returns within limits.

3.2.14 Compatibility.- The form factor of the telemetry transmitting set shall be compatible with that specified in paragraph 3.4 A reduction in size is permissible provided an adaptor plate is supplied to match the mounting provisions, and the connectors as specified are used.



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3.2.15 Enclosure.- The telemetry transmitting set shall be completely enclosed in a case which as a minimum shall function as a dust cover and provide protection for the installed components against hard handling.

3.3 Performance.-

3.3.1 General.- Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Antenna System Compatibility.- The telemetering set shall operate satisfactorily with a minimum output of 20 watts when connected to an antenna system having a VSWR from unity to 2:1 (leading or lagging phase angles). Operation into a VSWR worse than 3:1 for an indefinite period or into an open circuit or short circuit for a period up to (5) five minutes shall not be harmful to the power amplifier stage. No arcing, excessive frequency deviation with spurious signal radiations, output power less than specified minimums or other malfunctions will be considered satisfactory performance.

3.3.3 Frequency Range.- The transmitter carrier frequency range shall be 225 to 260 MHz. Specific frequencies shall be specified by the procuring agency. The components and crystal that have to be replaced to change the frequency shall be clearly identified in the assembly. Removal and substitution of these items shall not be impeded by their position within the assembly or by other components or structures. No special equipment, tools or fixtures shall be required to effect this frequency change.

3.3.4 Frequency Stability.- The transmitter carrier frequency shall not vary more then plus or minus 0.01 per cent of the carrier frequency as specified in specification MIL-STD-442 when 100 per cent modulated by subcarriers.

3.3.5 Frequency Deviation.- The carrier frequency deviation shall be ±125KC ±25KC.

3.3.6 Output Power.- The transmitting set output power shall be 25 watts minimum delivered into a 50-ohm resistive load under standard conditions and shall remain within ±1.5db under environmental conditions set forth in 3.2.11, and with input power conditions of +28 ±10% volts D.C. See paragraph 3.3.2 for load mismatch requirements.



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3.3.7 Modulation.- The RF carrier shall be modulated by ten (10) subcarrier oscillators. The carrier deviations produced by the individual subcarriers shall be the optimum values compatible with good signal to noise ratios and low inter-modulation distortion. The carrier modulation pre-emphasis schedule shall be furnished by the manufacturer.

3.3.7.1 Noise, Intermodulation or Crosstalk.- The telemetry system shall not generate noise, intermodulation or crosstalk on the demodulated output of each channel in excess of that caused by placing an equivalent peak-to-peak signal of two (2) per cent of the rated maximum input voltage on the input of a corresponding channel. This shall be defined as follows:

$$\text{IM (per cent Intermodulation or Crosstalk)} = \frac{Vpp}{Vdc} \times 100$$

Where:

Vpp = Peak-to-peak ripple voltage present on the output of a discriminator at a receiving station as seen on an oscilloscope.

Vdc = Voltage output of a discriminator corresponding to the full design bandwidth (DBW) deviation of a specific subcarrier channel.

3.3.8 Subcarrier Modulation.- All subcarrier oscillators shall be capable of being frequency modulated plus and minus 7.5 per cent of the subcarrier center frequency by the intelligence signal (0 to 5 volts D.C.).

3.3.9 Subcarrier Bandpass.- Each subcarrier bandpass shall provide the channel frequency response compatible with the subcarrier deviation ratio and optimum signal-to-noise ratio as specified in specification MIL-STD-442.

3.3.10 Subcarrier Frequencies.- The transmitting set shall operate satisfactorily when the following subcarrier oscillators are employed:



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| IRIG Band | Center Frequency |
|---|---|
| 5 | 1.3 KC |
| 6 | 1.7 KC |
| 7 | 2.3 KC |
| 8 | 3.0 KC |
| 9 | 3.9 KC |
| 10 | 5.4 KC |
| 11 | 7.35 KC |
| 12 | 10.5 KC |
| 13 | 14.5 KC |
| 14 | 22.0 KC |

3.4 Detail Requirements.-

3.4.1 Telemetry Transmitting Set.- The telemetry transmitting set shall meet the following requirements:

3.4.1.1 Function.- The telemetry transmitting set shall contain all of the components necessary for the FM/FM transmission of the target information.

3.4.1.2 Form Factor.- The telemetry transmitting set shall meet the materials, bonding, shielding and performance requirements of specification MIL-C-172, and shall not exceed 8 inches in width by 13 inches in length by 5.4 inches in height. The dimensions and mounting provisions are shown in figure 1. A reduction in package size is permissible provided an adapter plate is supplied to match the mounting provisions specified in paragraph 3.2.14.

3.4.1.3 Weight.- The weight of the entire telemetry transmitting set shall not exceed 13 pounds in the ten-channel configuration.

3.4.1.4 Content of the Unit.- The telemetry transmitting set shall include the following components: RF Power Amplifier, RF Transmitter, Subcarrier Oscillators, Mixer Amplifier, Transducer Power Supply, Calibration Network, Multiplexing Network and Interconnecting Wiring. An Isolator may be required to meet the antenna system mismatch requirements of this specification. Heaters may be required to meet the low temperature specification performance requirements.



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3.4.1.5 Controls and Displays.- All controls and lights must be externally accessible and located on the same surface as the external connectors. The following shall be provided:

| | |
|---|---|
| RF Power Switch | A switch permitting control of the power supply for the RF transmitter and power amplifier. Provisions shall be incorporated to permit safety wiring of the R.F. power switch in the "on" position. |
| Calibration Switch | A pushbutton switch to control the three-point calibration of the subcarrier oscillators. |
| Indicator Lights | A light to display the status of the calibration network. |

3.4.1.6 Electrical Connections.- Connections to external circuits shall be provided as follows:

| Reference Designation | Receptacle Type | Pin No. and Function |
|---|---|---|
| J101 | UG-1094/U | RF Output |
| J102 (input) | MS3114E-14-18P or Bendix PT07E-14-18P or equivalent. | A-Channel 11 Signal Input<br>B-Channel 12 Signal Input<br>C-Channel 13 Signal Input<br>D-Channel 14 Signal Input<br>E-Channel 5 Signal Input<br>F-Channel 6 Signal Input<br>G-Channel 7 Signal Input<br>H-Channel 8 Signal Input<br>J-Channel 9 Signal Input<br>K-Channel 10 Signal Input |



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| Reference Designation | Receptacle Type | Pin No. and Function |
|---|---|---|
| J102 (Input) | MS3114E-14-18P<br>or Bendix<br>PT07E-14-18P<br>or equivalent | L-Ground<br>M-Ground<br>N-Ground<br>P-28 vdc Input<br>R-5 vdc Output<br>S-Spare<br>T-Calibrate Command<br>U-28 vdc Input |
| J103 (Test) | MS3114E-14-18S<br>or Bendix<br>PT07E-14-18S<br>or equivalent | A-Channel 5 Signal Output<br>B-Channel 6 Signal Output<br>C-Channel 7 Signal Output<br>D-Channel 8 Signal Output<br>E-Channel 9 Signal Output<br>F-Channel 10 Signal Output<br>G-Channel 11 Signal Output<br>H-Channel 12 Signal Output<br>J-Channel 13 Signal Output<br>K-Channel 14 Signal Output<br>L-28 vdc<br>M-5 vdc<br>N-Spare<br>P-Spare<br>R-Multiplexed Signal<br>S-Calibrator Output<br>T-Spare<br>U-Ground |
| J1 through J10 | Cannon DE-9S<br>or equivalent | Subcarrier Oscillator Power Signal Receptacles (see figure 2 for mechanical specifications)<br><br>1-Signal Input<br>2-Ground<br>3-Spare<br>4-Output Monitor<br>5-Signal Output<br>6-Chassis Ground<br>7-+28 vdc<br>8-Spare<br>9-Spare |



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3.4.2 RF Power Amplifier.- The power amplifier unit shall be fully solid state design and meet the following requirements.

3.4.2.1 Function.- The RF power amplifier increases the amount of radio frequency energy available to the transmitting antenna.

3.4.2.2 Contents of Unit.- The power amplifier shall contain the circuitry necessary to meet the following requirements.

| | |
|---|---|
| Power Output | Sufficient to provide a system output of 25 watts minimum under standard conditions and shall remain within ±1.5 db under environmental conditions set forth in 3.2.11, and with input power conditions of +28 vdc ±10%. See paragraph 3.3.2 for load mismatch requirements. |
| Output Impedance | 50 ohms |
| Input Impedance | 50 ohms |
| Frequency Range | 225 to 260 MC, field tuneable |
| Driving Power | 2 watts minimum |
| Bandwidth | 300 KC minimum |

3.4.2.3 Controls.- The power amplifier shall contain screwdriver adjustments for input and output tuning.

3.4.2.4 Electrical Connections.- Connections to external circuits shall be provided.

3.4.3 RF Transmitter.- The RF transmitter unit shall be fully solid state design and meet the following requirements.

3.4.3.1 Function.- The RF transmitter is the radio frequency generating unit and is frequency-modulated by subcarrier oscillators.

3.4.3.2 Contents of Unit.- The transmitter shall contain the circuitry necessary to meet the following requirements.

| | |
|---|---|
| Power Output | 2 watts minimum and shall remain within ±1.0db under the environmental conditions set forth in 3.2.11 and with input power conditions of 28 vdc ±10%. |



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| | |
|---|---|
| Output Impedance | 50 ohms |
| Frequency Range | 225 to 260 MC, field tuneable with crystal change |
| Frequency Stability | ±0.01 per cent |
| Modulation | True FM |
| Deviation | Can be deviated to ±150KC maximum |
| Modulation Frequency Range | 20 cps to 100 KC, ±3 db response |
| Spurious Modulation | Incidental FM less than ±5 KC |
| Modulation Sensitivity | 0.01v rms per 1 KC deviation |
| Modulation Distortion | 2 per cent maximum for 30 KC deviation at any subcarrier frequency |
| Radio Frequency Interference | As necessary to comply with specification MIL-I-6181 |

3.4.3.3 Controls.- The RF transmitter shall contain the necessary screwdriver adjustments for transmitter tuning.

3.4.3.4 Electrical Connectors.- Connections to external circuits shall be provided.

3.4.4 Subcarrier Oscillator.- Each subcarrier oscillator shall be solid state and shall meet the following requirements:

3.4.4.1 Functions.- The subcarrier oscillator takes a D.C. voltage input and transforms it into an A.C. voltage output.

3.4.4.2 Form Factor.- The subcarrier oscillator shall not exceed the size and mounting dimensions shown in figure 3.

3.4.4.3 Contents of Unit.- The subcarrier oscillator shall contain the circuitry necessary to meet the following requirements.



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| | |
|---|---|
| Input, Instrumentation | 0 to 5 volts D.C. |
| Input Impedance | 500K ohms minimum |
| Linearity | ±0.3 per cent DBW deviation from the straight line drawn between the upper and lower limits of the oscillator (IRIG Band edges) |
| Drift | After 10 minute operation, drift for the next 8-hour period at 25 degrees C is to be less than 0.5 per cent DBW |
| Output Impedance | 50K ±20% |
| Output Voltage | 75 MV to 850 MV P-P into a 10K load |
| Distortion | Less than 1.0 per cent |
| Amplitude Modulation | Less than ±5 per cent |
| Stability | Less than ±1 per cent DBW shift in frequency for a ±10 per cent change in supply voltage |
| Phasing | Increasing input voltage produces increasing output frequency |
| Center Frequency Stability | ±2 per cent DBW from that measured at 25 degrees C over the environmental conditions set forth in 3.2.11 |
| Input Voltage | 28 ±10% volts D.C. |



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3.4.4.4 Controls.- The subcarrier oscillator shall contain the necessary screwdriver adjustments for setting the frequency limits and the output voltage level. Note that in the initial system setup, bandedge adjustments to within ±3% DBW of the ideal shall be provided.

3.4.4.5 Electrical Connections.- Connections to external circuits shall be provided as follows:

| Receptacle Type | Pin No. and Function |
|---|---|
| Cannon DE-9P or equivalent | 1. Signal Input<br>2. Ground<br>3. Spare<br>4. Output Monitor<br>5. Signal Output<br>6. Chassis Ground<br>7. +28 vdc<br>8. Spare<br>9. Spare |

3.4.5 Transducer Power Supply.- The transducer power supply shall be solid state design and meet the following requirements:

3.4.5.1 Function.- The transducer power supply converts the 28 vdc supply voltage into a 5-volt D.C. regulated voltage.

3.4.5.2 Contents of Unit.- The transducer power supply shall contain the solid state circuitry necessary to meet the following requirements.

| | |
|---|---|
| Output Voltage | 5.0 vdc adjustable ±.2 volts with a tolerance of ±0.01 volt |
| Load Current | 50 milliamperes maximum |
| Regulation | ±0.5 per cent for any changes in load, line voltage or environmental conditions required by this specification |
| Ripple Voltage | 10 millivolts peak-to-peak |
| Short Circuit Protection | As necessary to prevent damage due to short circuit of output terminals |
| Input Voltage | 28 vdc ±10 per cent |



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3.4.5.3 Controls.- The transducer power supply shall contain a screwdriver adjustment for the output voltage.

3.4.5.4 Electrical Connections.- Connections to external circuits shall be provided.

3.4.6 Calibration Network.- The calibration network shall meet the following requirements:

| | |
|---|---|
| Calibration Steps | Three accurate voltage levels applied to the subcarrier oscillator inputs. The calibration voltage shall be obtained from the transducer Power Supply. |
| Calibration Accuracy | ±0.3 per cent of full scale. |
| Calibration Duration | Duration of each calibration voltage level to be indefinite. Each calibration level to be stepped upon remote and manual commands. |
| Calibration Trigger | 22-32 vdc pulse to be required to step each calibration level. |
| Calibration Channels | 1 to 10 subcarrier oscillators may be calibrated simultaneously. |
| Calibration Modes | The trigger pulse (may be DC to 50m sec. pulse) shall step the calibrator circuit to the next step where it will remain until another step is applied as follows:<br><br>Step 1. VCO input goes to +5 vdc and remains whether the trigger is held or removed.<br><br>Step 2. VCO input goes to +2.5 vdc and remains, etc.<br><br>Step 3. VCO input goes to zero and remains, etc.<br><br>Step 4. VCO input goes to data position and remains, etc. |



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3.4.6.1 Electrical Connections.- Connections to external circuits shall be provided.

3.4.7 Mixer-Amplifier.- The mixer-amplifier shall be solid state and shall meet the following requirements.

3.4.7.1 Function.- The mixer-amplifier shall provide sufficient amplification to properly deviate the RF transmitter.

3.4.7.2 Form Factor.- The mixer-amplifier shall be identical with the subcarrier oscillator shown in figure 3.

3.4.7.3 Contents of Unit.- The mixer-amplifier shall contain the solid-state circuitry necessary to meet the following requirements:

| | |
|---|---|
| Frequency Response | ±0.5db from 100 to 100,000 cps |
| Input Impedance | 10,000 ohms |
| Output Impedance | 1,000 ohms maximum |
| Input Signal | 100 millivolts minimum |
| Output Signal | 2.0 volts rms |
| Harmonic Distortion | 0.5 per cent maximum |
| Gain | 0.5 to 20 adjustable |
| Noise | Less than 10 mv. peak-to-peak at a gain of 20 |
| Stability | Less than ±2.5 per cent change under all conditions required by this specification |
| Input Voltage | 28 vdc ±10 per cent |

3.4.7.4 Controls.- The mixer-amplifier shall contain a screwdriver adjustment for setting the gain.

3.4.7.5 Electrical Connections.- Connections to external circuits shall be provided as follows:



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| Receptacle Type | Pin No. and Function |
|---|---|
| Cannon DE-9P or equivalent | 1. Spare<br>2. Signal Ground<br>3. Spare<br>4. Spare<br>5. Signal Input<br>6. Chassis Ground<br>7. +28 vdc<br>8. Spare<br>9. Signal Output |

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements. The contractor shall submit, prior to start of manufacture, test procedures for preproduction and acceptance inspection prepared in accordance with MIL-T-18303A. (See 6.6.3)

(a) Preproduction Tests

(b) Production Sample Tests

(c) Acceptance Tests

4.1.2 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests.- Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of Contractor's Demonstration Tests.

4.2.1 Contractor's Demonstration Tests.- Contractor's demonstration tests shall be accomplished under the responsibility of the contractor, and shall be conducted in accordance with the approved test procedure. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired.



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Contractor's not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Scope of Tests.- Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of the specification and the contract. Preproduction tests shall include the following tests:

(a) Environmental Tests

(b) Radio Interference Tests

(c) Reliability Tests

(d) Frequency Range Test

4.2.2.1 Environmental Tests.- Environmental tests shall be performed in accordance with specification MIL-T-5422. Paragraph 3.2.11 details conditions.

4.2.2.2 Radio Interference Tests.- Radio interference tests shall be in accordance with specification MIL-I-6181. In addition, spurious and harmonic output tests shall be performed with VSWR at 2:1 with the phase angle varied from 0 to 360 degrees.

4.2.2.3 Reliability Tests.- Reliability tests shall be in accordance with specification MIL-R-22973 with the following additions and exceptions.

4.2.2.3.1 Quantity of equipment to be tested.- One preproduction prototype of the equipment shall be tested.

4.2.2.3.2 Length of Tests.- The total unit hours of test, not including any burn-in period, shall be 270 hours equipment on time.

4.2.2.3.3 Test Level.-

| | |
|---|---|
| Temperature | -54°C to +55°C |
| Vibration | MIL-E-5400, Curve I, Figure 2 |
| Heating Period | 4.9.hours |
| Cooling Period | Time to stabilize at low temperature, 1.1 hours |
| Input Voltage | Maximum specified voltage at maximum temperature. Minimum specified voltage at minimum temperature. |



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4.2.2.3.4 Duty Cycle.- The heating period shall be 4.9 hours in accordance with the chart of Figure 4. The equipment shall be off during the cooling period and operating during the heating period. Seven periods of vibration equally spaced throughout the heating period, shall be imposed during each duty cycle. Each period of vibration shall consist of a vibration cycle scanning from 5 to 500 cps and back to 5 cps in about 15 minutes. During vibration the equipment shall be mounted solidly, that is without shock vibration mounts. The equipment shall be vibrated for an equal number of cycles in each of the three principle axes.

| Temperature Condition | Equipment | Time |
|---|---|---|
| Ambient to -54°C | Non-operating | 0.6 hr. |
| -54°C to -20°C | Operating | 0.5 hr. |
| -20°C to +55°C | Operating | 1.4 hr. |
| +55°C | Operating | 3.0 hr. |
| +55°C to Ambient | Non-operating | 0.5 hr. (or greater) |

4.2.2.3.5 Performance Characteristics.- The equipment shall comply with the performance characteristics of 3.3 and 3.4 of this specification.

4.2.2.3.6 Equipment Failure.- The criteria for equipment failure shall be as defined in paragraph 3.12 of MIL-R-22973.

4.2.2.3.7 Maintenance.- Maintenance shall be limited to that specified in paragraph 3.9 of MIL-R-22973. Preventative maintenance shall not be performed. The failure of VCO's, Mixer Amp. or Transducer Power Supply may be corrected by their replacement. Failure of the Transmitter, Power Amp. or components of the main frame shall be corrected by repair.

4.2.2.3.8 Replacement Parts.- The replacement of parts shall be governed by paragraph 3.10 of MIL-R-22973.

4.2.2.3.9 Disposition of Equipment.- The disposition of equipment shall be specified by the procuring activity.

4.2.2.4 Frequency Range Test.- A test demonstrating the ability to change the transmitter frequency over the frequency range shall be conducted as early as possible in the Preproduction Test program. This test need not be conducted on a complete equipment if the affected components can be easily removed from the set. The components must be identical to production items in design and construction (potting, identification of parts; etc.).



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4.2.3 Preproduction Approval.- Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk.

4.3 Production Sample Tests.- When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests.- This equipment may be subjected to any and all tests the procuring activity deems necessary to ensure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, performance, and reliability, and that it meets all applicable requirements.

4.3.2 Production Sample Approval.- Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample.- On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor may resubmit the equipment for acceptance and subsequent shipment if so directed by the procuring agency.

4.4 Acceptance Tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests
(b) Reliability Assurance Tests



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4.4.1 Individual Tests.- Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational Test

4.4.1.1 Examination of Product.- Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test.- Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.2. Reliability Assurance Tests.- Shall be conducted on production equipments which have successfully passed the individual tests.

4.4.2.1 Scope of Tests.- Reliability assurance tests shall be conducted in accordance with MIL-STD-781, and paragraph 4.4.2.3 Duty Cycle of this specification.

4.4.2.1.1 Quantity to be Tested.- The first eight (8) production units shall be tested for the Qualification Plan; thereafter the quantity shall be in accordance with paragraph 4.2.4.1 of MIL-STD-781 except that for production rates of less than ten (10) units per month, a production lot shall consist of two months production.

4.4.2.1.2 Test Level.-

| | |
|---|---|
| Temperature | -54°C to +55°C |
| Vibration | MIL-E-5400, Curve II, Figure 2 |
| Heating Period | 4.9 hours |
| Cooling Period | Time to stabilize at low temperature |
| Input Voltage | Maximum specified voltage at maximum temperature. Minimum specified voltage at minimum temperature. |



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4.4.2.2 Accept-Reject Criteria.- Shall be as specified in MIL-STD-781, Accept-Reject Criteria, paragraph 4.2.8.1.

4.4.2.3 Duty Cycle.- The heating period shall be 4.9 hours. Five periods of vibration equally spaced throughout the heating period shall be imposed during each cycle. The first period of vibration starting at the end of the 20 minute warm-up time and the last period of vibration ending with the conclusion of the heating period. A performance check shall be conducted during the first and last periods of vibration and the data recorded. Each equipment shall be vibrated in the principle axis which resulted in the largest number of failures during reliability testing of MIL-R-22973.

| Temperature | Equipment | Time |
|---|---|---|
| Ambient to -54°C | Off | 0.6 hr. |
| -54°C to -20°C | On | 0.5 hr. |
| -20°C to +55°C | On | 1.4 hr. |
| +55°C | On | 3.0 hr. |
| +55°C to Ambient | Off | 0.5 hr. or greater |

A Thermal Survey shall be conducted; or data from a previous Thermal Survey, conducted on the Telemetry Transmitting Set, may be accepted. The chamber may be taken from -20°C to +55°C as fast as practical. However, the "hot spot" as determined by the Thermal Survey will determine the 1.4 hours to stabilize as defined in paragraph 5.1.5 of MIL-STD-781.

4.4.2.4 Performance Characteristics.- The equipment shall comply with the performance characteristics of paragraphs 3.3 and 3.4 of this specification.

4.4.2.5 Equipment Failure.- The criteria for equipment failure shall be as defined in paragraph 5.5 of MIL-STD-781.

4.4.2.6 Maintenance.- Maintenance shall be limited to that specified in paragraph 5.8 of MIL-STD-781. Preventative maintenance shall not be performed.

4.4.2.7 Replacement Parts.- The replacement of parts shall be governed by paragraph 5.9 of MIL-STD-781.

4.4.2.8 Disposition of Equipment.- The disposition of equipment shall be specified by the procuring activity.



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4.4.3 Equipment Failure.- Should a failure occur during either the sampling or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Where practicable, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Presubmission Testing.- No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.6 Rejection and Retest.- Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.7 Equipment Changes.- After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.- All major units and parts of equipment shall be packaged level A or C, and packed Level A, B or C as specified in the contract or order (see 6.6). Levels of packaging and packing and marking shall be in accordance with specification MIL-E-17555.

6. NOTES

6.1 Use.- The equipment specified herein will be installed in a QT-33 or BMQ-34 A & E radio controlled target. This telemetering set shall be capable of using the existing transducers and antenna of these targets. These items are not a part of this specification.

6.2 Test Values.- Normal and limiting values of performance data shall be determined at input voltages of 28 ±10% volts D.C. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.



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6.3 Precedence of Documents.- When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract. The contract shall have precedence over any specification.

(b) This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives.- Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Accessory Material.-

6.5.1 Preproduction Model Material.- The contractor shall furnish, in addition to the complete equipment furnished for preproduction tests, the following:

(a) Complete set of interconnecting cables

(b) One set of spare parts peculiar to the equipment (standard parts such as resistors, capacitors, and tubes in common use are not required).



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6.6 Ordering Data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable level of packaging and packing required (see 5.1).

(c) Data required for operational use and for maintenance and procurement purposes.

6.6.1 Design Data.- The contractor should furnish only those items of the following data that are specified in the contract or order.

(a) Three copies of outline dimensional sketches of all major and minor assemblies of any detail parts not internal therein, showing projections.

(b) Three copies of a practical wiring diagram of each major assembly showing the physical location and connections of detail parts and sub-assemblies with reference symbols and terminal numbers indicated.

(c) Three copies of brief operation instructions.

(d) Prior to shipping the preproduction sample equipment to the specified destination for Service Approval Tests, three copies of a report presenting data obtained by the contractor in conducting the Contractor's Demonstration Tests (para. 4.2.1) should be submitted to the procuring activity for review and approval. Included in this report should be an analysis of all failures which occurred, and suggestions for improvements in design which might be incorporated in later productions.

(e) In addition to the complete equipment submitted for service approval tests, the contractor should also submit the accessory material and design and test data specified in specification MIL-E-5400. This information should indicate the physical and electrical characteristics of the equipment, and establish the equipment's compliance with applicable requirements.



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(f) Three copies of a complete schematic diagram reduced to its simplest form, showing the circuits of all assemblies in schematic form with electrical interconnection indicated.

6.6.2 Acceptance Test Data.- The contractor should furnish test reports showing quantitative results of all Acceptance Tests (para. 4.4). Such reports should be signed by an authorized representative of the contractor or laboratory, as applicable.

6.6.2.1 Special Test Procedures.- Where previously approved test procedures for Preproduction and Sampling tests are not applicable to required Special Tests, the contractor should prepare a test procedure and submit it to the procuring activity for approval prior to conducting such tests.

6.6.2.2 Equipment Failure.- Should a failure occur during either the sampling or special tests, the contractor shall submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests (see 4.4.4).

6.6.3 Test Procedures.- The procedures used for conducting preproduction tests, production sample tests and acceptance tests should be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract. Specification MIL-T-18303A should be used as a guide for the preparation of test procedures. When test procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

6.6.4 Rejection and Retest.- Before resubmitting equipment which was rejected and subsequently reworked per paragraph 4.6, full particulars concerning previous rejection and the action taken to correct the defects found in the original should be furnished the government inspector.

6.6.5 Non-Standard Parts and Materials.- Approval for the use of non-standard parts and materials referred to in paragraph 3.1.1 should be obtained as outlined in specification MIL-D-18300.

6.6.6 Nomenclature, Nameplates and Serial Numbers.- The contractor shall request assignment of nomenclature-type designations, serial number and nameplate approval to identify the equipment and major units to be furnished under the contract in accordance with specification MIL-N-18307.



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6.6.7 Preproduction Test Reports.- Following the preproduction tests, three copies of a report presenting data obtained by the contractor in conducting the Preproduction Tests (4.2.3) should be submitted to the procuring activity for review and approval. Included in this report should be an analysis of all failures which occurred, and suggestions for improvements in design which might be incorporated in later productions. The Preproduction Test Reports shall report on all of the appropriate tests referenced under paragraph 4.2.3 and following subparagraphs.

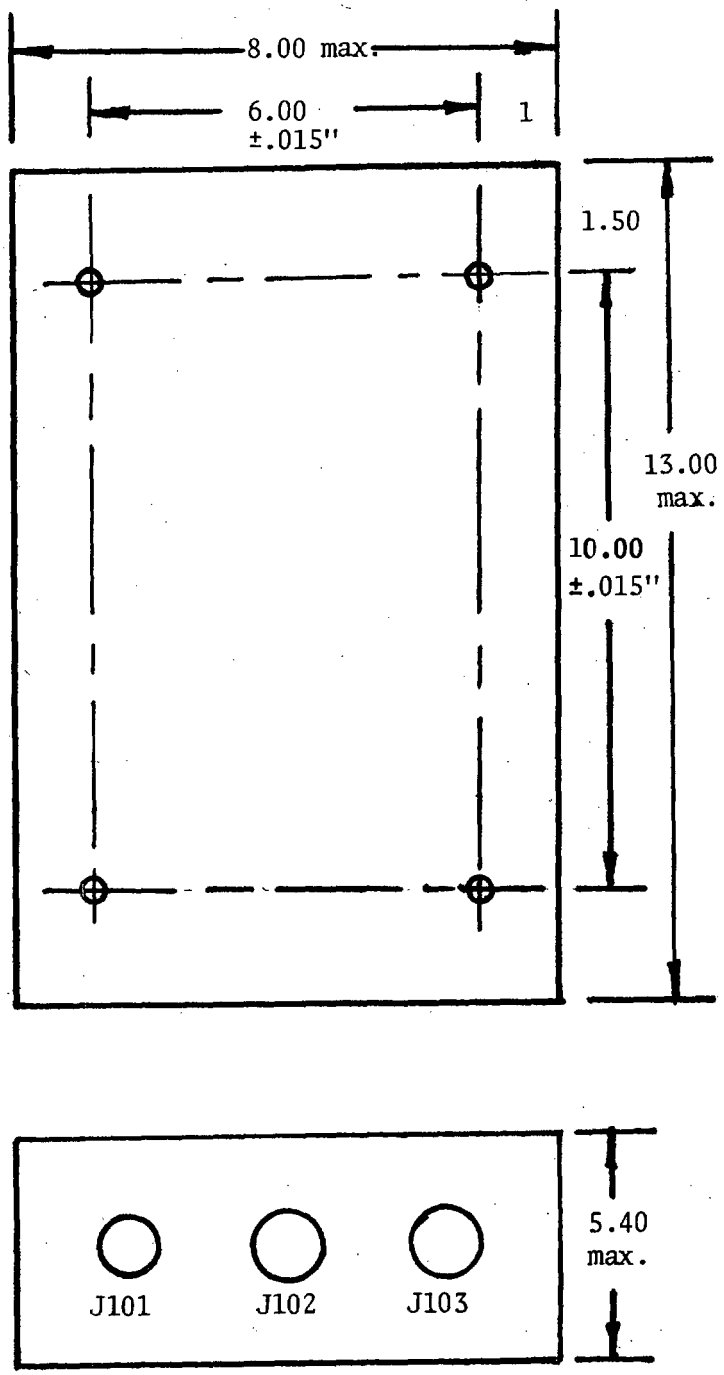
6.6.8 Equipment Changes.- After approval of the preproduction model, any changes which affect weight, installation interchangeability, and interchangeability of maintenance parts of assemblies should require prior approval of the procuring activity. To obtain procuring activity approval, the contractor should submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Avionics Change (AVC) and an Avionics Bulletin (AVB) to cover the approved change. Engineering Change Proposals, Avionics Changes, and Avionics Bulletins should be prepared in accordance with MIL-T-23336. ECP's should be prepared in accordance with MIL-STD-480.

Project No. 5821-N086



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Inserts (1/4-28)
.36 Min. Deep
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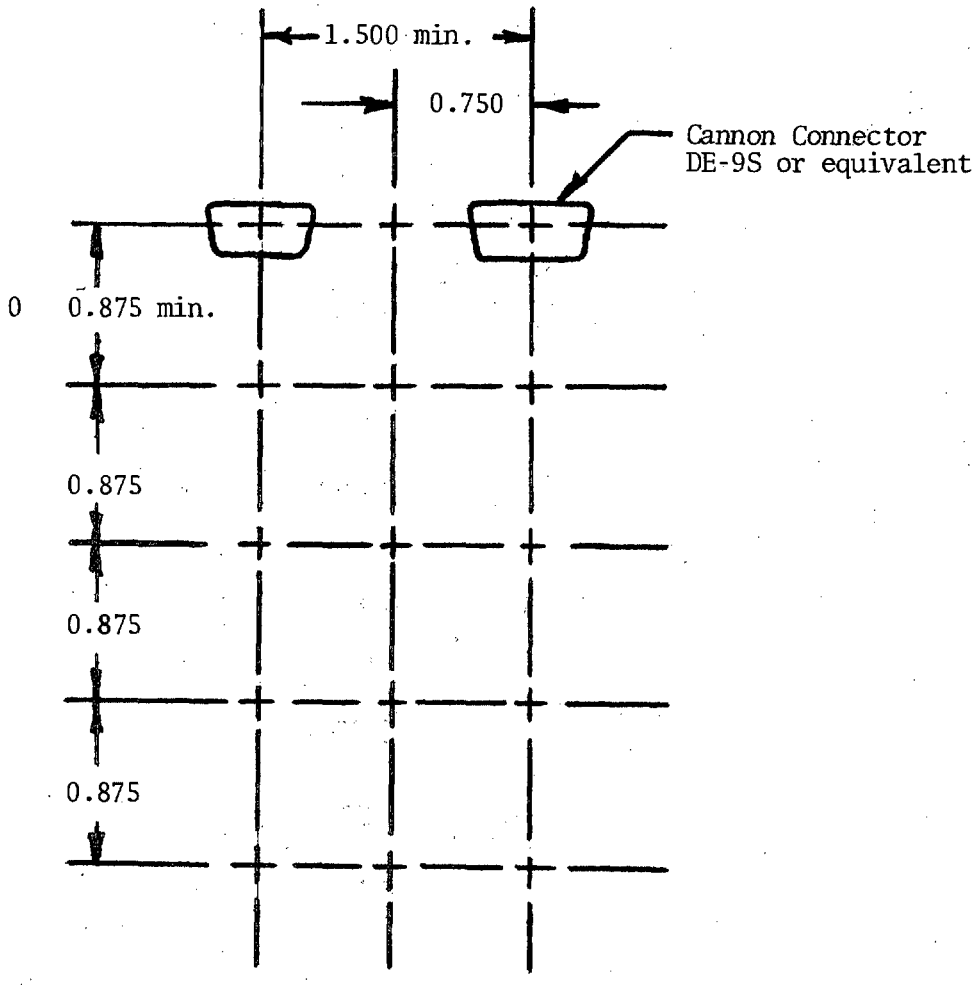



Telemetry Transmitting Set

Figure 1



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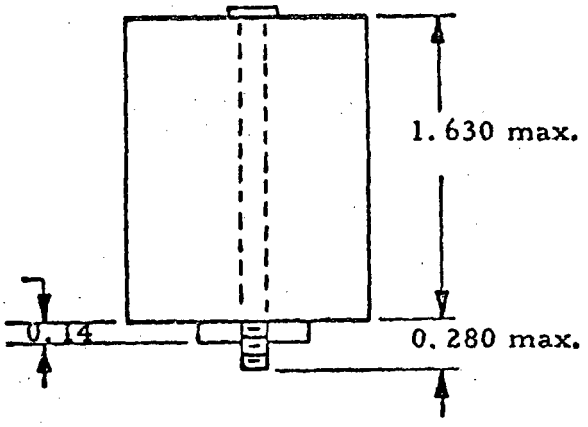

Dimensions in Inches

Subcarrier Oscillator Connector Spacing Requirements

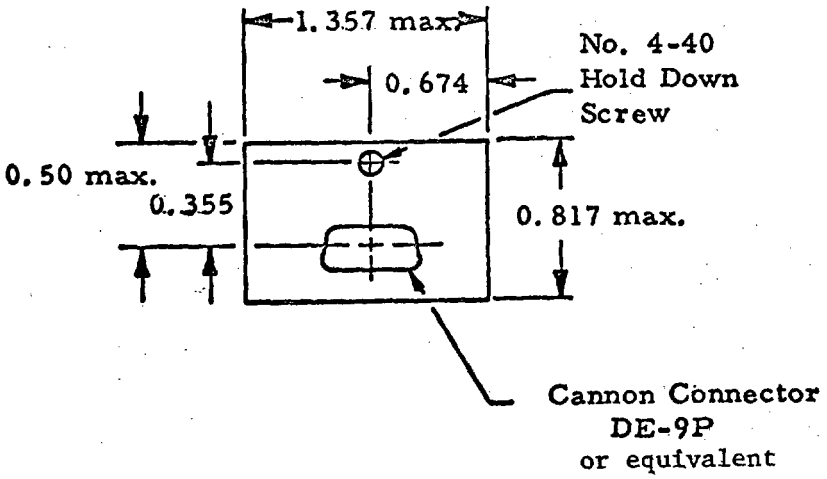
Figure 2



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Dimensions in Inches

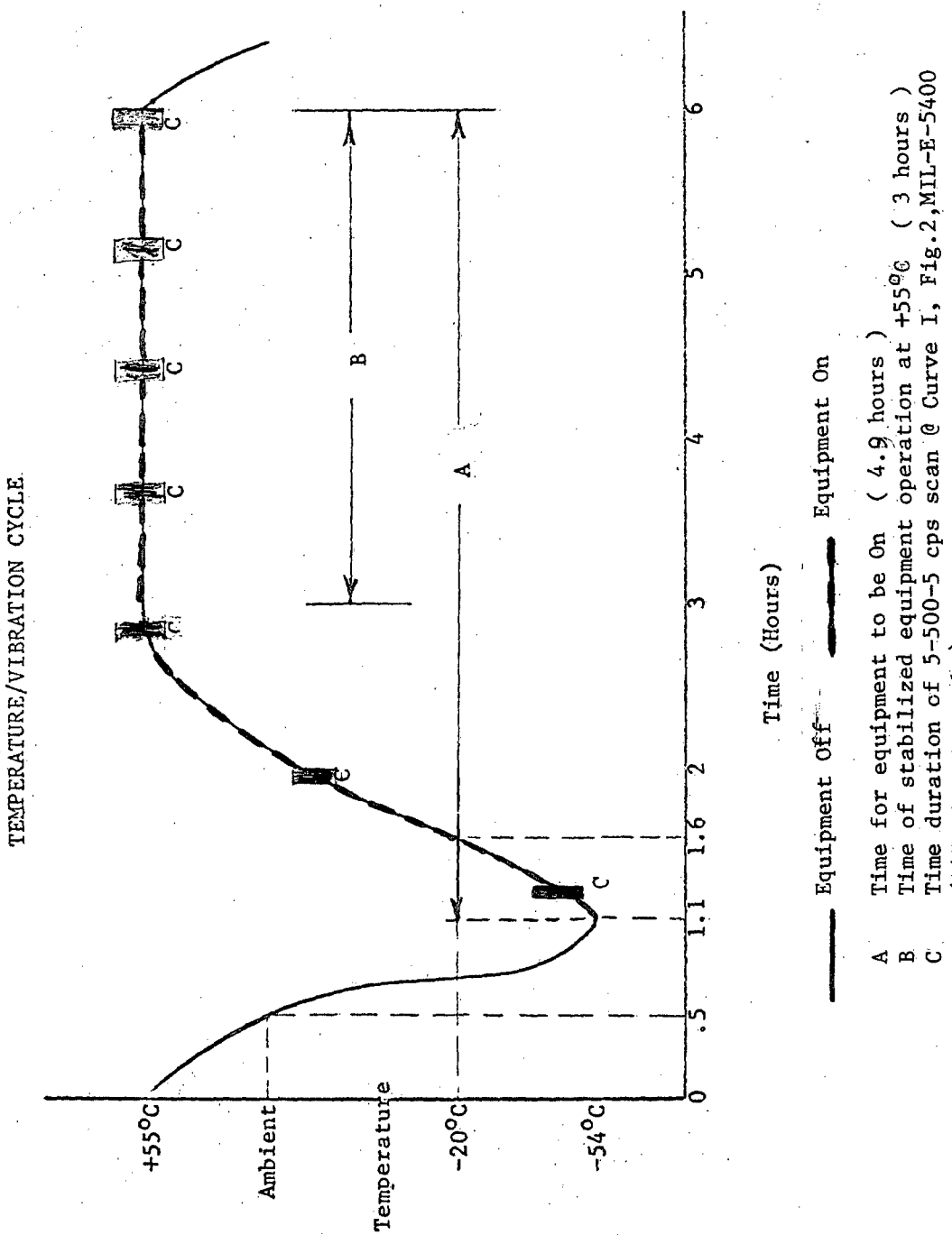



Subcarrier Oscillator Outline Drawing

Figure 3



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RELIABILITY ASSURANCE ENVIRONMENTAL
TEST TIME PROFILE

**Figure 4**

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# SPECIFICATION ANALYSIS SHEET.

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-24015B(AS) TRANSMITTING SET, TELEMETRY DATA AN/AKT-21

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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